Abdellah Fourtassi

Associate Professor (Maître de Conférences HDR) in Computer Science
Aix-Marseille Université · LIS / TALEP · Institute of Language, Communication & the Brain (ILCB)
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RESEARCH FOCUS: How interaction shapes language learning in humans and machines. I build models that learn from multimodal interaction under cognitively plausible constraints, scaling learning theories to ecologically realistic environments.

ACADEMIC APPOINTMENTS & ROLES

Aix-Marseille Université - Associate Professor (Maître de Conférences HDR), Computer Science	**2025-present**
Aix-Marseille Université - Assistant Professor (Maître de Conférences), Computer Science	**2019-2025**
Director, Computation in Communicative Development (CoCoDev) team	**2019-present**
Deputy Director for Research, Institute of Language, Communication & the Brain (ILCB)	**2024-present**
Co-lead, *NLP & Cognition* national working group, CNRS GDR TAL	**2025-present**
Co-Director, Master in Cognitive Science (MASCO), Aix-Marseille Université	**2020-2021**
Associate Editor, *Dialogue & Discourse*	**2026-present**
Visiting Researcher, INRIA Paris (Délégation Recherche)	**2021-2022**
Postdoctoral Research Fellow, Department of Psychology, Stanford University	**2016-2019**

EDUCATION

PhD in Cognitive Science, École Normale Supérieure (ENS), Paris - summa cum laude	**2015**
Master's in Cognitive Science (CogMaster), EHESS, Paris	**2012**
Science & Engineering Diploma, École Polytechnique, Paris	**2012**
CPGE/MP*, Mathematics & Physics, Lycée Moulay Youssef, Rabat	**2006-2008**

RESEARCH FUNDING

As Principle Investigator

ANR PRME - €380k **2026-2030**
Grounded Language Learnability (GLAD)

ANR JCJC - €287k **2022-2025**
Mastering the Art of Conversation in Middle Childhood (MACoMiC)

As International Co-PI

Leverhulme Trust (UK), £290k (PI: G. Vigliocco, UCL) **2025-2028**
What Makes Social Interaction a Powerful learning Tool

SELECTED HONORS & DISTINCTIONS

- Keynote speaker: SIGDIAL (2025), Child Language Symposium (2026), LM Playschool Workshop at EMNLP (2026)
- PhD advisee Mitja Nikolaus received the Aix-Marseille Université Best PhD Thesis Award (2024)
- Marie Skłodowska-Curie European Fellowship, €185k (2019; declined upon accepting a faculty position at AMU)
- Stanford Data Science Seed Grant, $50k (2018-2019; conceived and led by A. Fourtassi; formal PI: M. Frank)
- Fyssen Foundation Postdoctoral Study Grant, €40k (2016-2018)
- France-Stanford Collaborative Award, $15k (2015-2016, conceived and led by A. Fourtassi; formal PIs: M. Frank & E. Dupoux)
- CoNLL Best Paper Award finalist, for the paper *Fourtassi & Dupoux (2014)*
- ACL Best Student Paper Award $1,000, for the paper *Fourtassi & Dupoux (2013)*

PUBLICATIONS

Book

1. **Fourtassi, A.** (forthcoming). *Language Development with Deep Learning*. Cambridge University Press.

Edited collections

2. Ma, MZ., Liu, E., Liu, J., Chang, TA., **Fourtassi, A.**, Warstadt,, A., Hahn M., Sun, W., Shi, F. (2026). Proceedings of the 1st Workshop on Computational Developmental Linguistics (CDL). In *ACL* (Annual Meeting of the Association for Computational Linguistics)

3. Agee, L., Monfils, M., & **Fourtassi, A.** (2025). Editorial: The Neural and Behavioral Methods of Social Learning. *Frontiers in Human Neuroscience.*

4. Apidianaki, M., **Fourtassi, A.**, & Padó, S. (2024). Editorial: Language Learning, Representation, and Processing in Humans and Machines. *Computational Linguistics.*

Journal articles

5. Goumri, D., Becerra-Bonache, L., & **Fourtassi, A.** (under review). The Interaction Engine Continues to Develop in Middle Childhood: Evidence from Gaze–Speech Coordination.
6. Brahimi, Y., & **Fourtassi, A.** (under review). Slow but Predictive: Turn-Taking Development in Child-Adult Conversation.
7. Agrawal, A., Favre, B., & **Fourtassi, A.** (2026). Scaffolding Early Dialogue: A Unified Account of Response Contingency in Child-Caregiver Interaction. *Cognitive Science,* 50(7).
8. Nikolaus, M., & **Fourtassi, A.** (2026). Modeling Children's Grammar Learning via Caregiver Feedback in Natural Conversations. *Philosophical Transactions of the Royal Society B,* 381.
9. Bodur, K., **Fourtassi, A.**, Fredouille, C., & Meunier, C. (2026). Speech Reduction in Middle Childhood: The Case of Child-Caregiver Interactions. *First Language.*
10. Gupta, S., Mazzocconi, C., & **Fourtassi, A.** (2025). Toward a child-centered, interactive approach to multimodal language development. *First Language,* 45(6).
11. Mazzocconi, C., O'Brien, B., Bodur, K., & **Fourtassi, A.** (2025). Do Children Laugh Like Their Parents? Conversational Laughter Mimicry and Acoustic Alignment in Middle Childhood. *Journal of Nonverbal Behavior,* 49.
12. Agee, L., **Fourtassi, A.**, & Monfils, M. (2025). Social Context as a Source of Variability in the Psychological Sciences. *Frontiers in Behavioral Neuroscience*, 18.
13. Cabiddu, F., Nikolaus, M., & **Fourtassi, A.** (2025). Comparing Children and Large Language Models in Word Sense Disambiguation: Insights and Challenges. *Language Development Research*, 5(1).
14. Bodur, K., Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2023). Using Video Calls to Study Children's Conversational Development: The Case of Backchannel Signaling. *Frontiers in Computer Science*, 5.
15. Wilson, K., Frank, M. C., & **Fourtassi, A.** (2023). Conceptual Hierarchy in Child-Directed Speech: Implicit Cues Are More Reliable. *Journal of Cognition & Development*, 24(4).
16. Nikolaus, M., & **Fourtassi, A.** (2023). Communicative Feedback in Language Acquisition. *New Ideas in Psychology*, 68, 100985.
17. Nikolaus, M., Maes, E., Auguste, J., Prévot, L., & **Fourtassi, A.** (2022). Large-Scale Study of Speech Acts' Development in Early Childhood. *Language Development Research*, 2(1).
18. Jiang, H., Frank, M. C., Kulkarni, V., & **Fourtassi, A.** (2022). Stability and Change in Caregivers' Word Usage Across Early Childhood. *Cognitive Science*, 46(7).
19. **Fourtassi, A.**, Regan, S., & Frank, M. C. (2021). Continuous Developmental Change Explains Discontinuities in Word Learning. *Developmental Science*, 24(2).
20. **Fourtassi, A.**, & Frank, M. C. (2020). How Optimal Is Word Identification Under Multimodal Uncertainty? *Cognition*, 199.
21. **Fourtassi, A.**, Bian, Y., & Frank, M. C. (2020). The Growth of Children's Semantic and Phonological Networks: Insights from 10 Languages. *Cognitive Science*, 44(7).

Peer-reviewed conference proceedings

22. Liu*, J., Schweitzer*, M., & **Fourtassi, A.** (2026). Which Forms of Caregiver Feedback Support Grammar Learning? A Reinforcement-Learning Study of Child-Like Language Models. *EMNLP* (Conference on Empirical Methods in Natural Language Processing).
23. Jumelet, J., **Fourtassi, A.**, Haga, A., Bunzeck, B., ... Choshen, L. (2026). BabyBabelLM: A Multilingual Benchmark of Developmentally Plausible Training Data. *EACL* (19th Conference of the European Chapter of the Association for Computational Linguistics).
24. Brahimi, Y., Blanc, C., & **Fourtassi, A.** (2026). Predicting Turn-Taking in Child-Adult Conversations Using Voice Activity Projection. *IWSDS* (16th International Workshop on Spoken Dialogue Systems Technology).
25. Liu, J., & **Fourtassi, A.** (2025). Benchmarking LLMs for Mimicking Child-Caregiver Language in Interaction. *SemDial* (29th Workshop on the Semantics and Pragmatics of Dialogue).
26. Agrawal, A., Favre, B., & **Fourtassi, A.** (2025). Identifying Repair Opportunities in Child-Caregiver Interactions. *SemDial* (29th Workshop on the Semantics and Pragmatics of Dialogue).
27. Agrawal, A., Nikolaus, M., Favre, B., & **Fourtassi, A.** (2024). Automatic Coding of Contingency in Child-Caregiver Conversations. *LREC-COLING* (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation).
28. Nikolaus, M., Agrawal, A., Kaklamanis, P., Warstadt, A., & **Fourtassi, A.** (2024). Automatic Annotation of Grammaticality in Child-Caregiver Conversations. *LREC-COLING* (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation).

29. Goumri, D., ..., & **Fourtassi, A.** (2024). A Developmental Corpus of Child-Caregiver Face-to-Face vs. Video-Call Conversations in Middle Childhood. *LREC-COLING* (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation).
30. Schweitzer, M., Montagnini, A., **Fourtassi, A.**, & Schatz, T. (2024). Multimodal Integration in Audio-Visual Speech Recognition. *NeurIPS* Behavioral Machine Learning Workshop.
31. Goumri, D.-E., Becerra, L., & **Fourtassi, A.** (2024). Child-Caregiver Gaze Dynamics in Naturalistic Face-to-Face Conversations. *CogSci* (46th Annual Meeting of the Cognitive Science Society).
32. Agrawal, A., Favre, B., & **Fourtassi, A.** (2024). Analyzing Communicative Intent Coordination in Child-Caregiver Interactions. *CogSci* (46th Annual Meeting of the Cognitive Science Society).
33. Peirolo, M., Xu, Z., & **Fourtassi, A.** (2024). Development of Flexible Role-Taking in Conversations Across Preschool. *CogSci* (46th Annual Meeting of the Cognitive Science Society).
34. Goumri, D., Janssoone, T., Becerra, L., & **Fourtassi, A.** (2023). Automatic Detection of Gaze and Smile in Children's Video Calls. *ICMI* Companion (International Conference on Multimodal Interaction).
35. Cabiddu, F., Nikolaus, M., & **Fourtassi, A.** (2023). Modeling Children's Word Sense Disambiguation with Large Language Models. *CogSci* (45th Annual Meeting of the Cognitive Science Society).
36. Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2023). Communicative Feedback Signals in Response to Children's Grammatical Errors. *CogSci* (45th Annual Meeting of the Cognitive Science Society).
37. Mazzocconi, C., O'Brien, B., El Haddad, K., Bodur, K., & **Fourtassi, A.** (2023). Mimicking and Non-Mimicking Laughter in Child-Caregiver Conversation. *CogSci* (45th Annual Meeting of the Cognitive Science Society).
38. Agrawal, A., Liu, J., Bodur, K., Favre, B., & **Fourtassi, A.** (2023). Development of Multimodal Turn Coordination in Conversations. *CogSci* (45th Annual Meeting of the Cognitive Science Society).
39. Tiran, T., Meewis, F., **Fourtassi, A.**, & Dautriche, I. (2023). Typology of Topological Relations Using Machine Translation. *CogSci* (45th Annual Meeting of the Cognitive Science Society).
40. Nikolaus, M., Salin, E., Ayache, S., **Fourtassi, A.**, & Favre, B. (2022). Do Vision-and-Language Transformers Learn Predicate-Noun Dependencies? *EMNLP* (Conference on Empirical Methods in Natural Language Processing).
41. Liu, J., Nikolaus, M., Bodur, K., & **Fourtassi, A.** (2022). Predicting Backchannel Signaling in Multimodal Child-Caregiver Conversation. *ICMI* Companion (International Conference on Multimodal Interaction).
42. Misiek, T., & **Fourtassi, A.** (2022). Caregivers Exaggerate Their Lexical Alignment to Young Children Across Several Cultures. *SemDial* (26th Workshop on the Semantics and Pragmatics of Dialogue).
43. Hallart, C., Peirolo, M., Xu, Z., & **Fourtassi, A.** (2022). Development of Dialogue Contingency in Child-Caregiver Interaction. *SemDial* (26th Workshop on the Semantics and Pragmatics of Dialogue).
44. Bodur, K., Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2022). Backchannel Behavior in Child-Caregiver Video Calls. *CogSci* (44th Annual Meeting of the Cognitive Science Society).
45. Nikolaus, M., Prévot, L., & **Fourtassi, A.** (2022). Communicative Feedback Supports Intelligible Speech in Childhood. *CogSci* (44th Annual Meeting of the Cognitive Science Society).
46. Nikolaus, M., & **Fourtassi, A.** (2021). Interaction Between Perception- and Production-Based Learning in Early Semantic Development. *CoNLL* (25th Conference on Computational Natural Language Learning).
47. Dary, F., Nasr, A., & **Fourtassi, A.** (2021). TALEP at *CMCL* 2021 Shared Task: Predicting Eye-Tracking Data (ranked 2 out of 30 submissions). *CMCL* (11th Workshop on Cognitive Modeling and Computational Linguistics).
48. Bodur, K., Nikolaus, M., Kassim, F., Prévot, L., & **Fourtassi, A.** (2021). ChiCo: A Multimodal Corpus for the Study of Child Conversation. *ICMI* Companion (International Conference on Multimodal Interaction).
49. Nikolaus, M., & **Fourtassi, A.** (2021). Evaluating the Acquisition of Semantic Knowledge from Cross-situational Learning in Artificial Neural Networks. *CMCL* (11th Workshop on Cognitive Modeling and Computational Linguistics).
50. Nikolaus, M., Maes, E., Auguste, J., Prévot, L., & **Fourtassi, A.** (2021). Large-scale Study of Speech Acts' Development Using Automatic Labeling. *CogSci* (43rd Annual Meeting of the Cognitive Science Society).
51. Nikolaus, M., Maes, E., & **Fourtassi, A.** (2021). Modeling Speech Act Development in Early Childhood: The Role of Frequency and Linguistic Cues. *CogSci* (43rd Annual Meeting of the Cognitive Science Society).
52. Dary, F., **Fourtassi, A.**, & Nasr, A. (2021). On the Role of Low-level Linguistic Tasks for Reading Time Prediction. *CogSci* (43rd Annual Meeting of the Cognitive Science Society).
53. Misiek, T., Favre, B., & **Fourtassi, A.** (2020). Development of Multi-level Linguistic Alignment in Child-Adult Conversations. *CMCL* (10th Workshop on Cognitive Modeling and Computational Linguistics).

54. **Fourtassi, A.** (2020). Word Co-Occurrence in Child-Directed Speech Predicts Children's Free Word Associations. *CMCL* (10th Workshop on Cognitive Modeling and Computational Linguistics).

55. **Fourtassi, A.**, Wilson, K., & Frank, M. C. (2020). Discovering Conceptual Hierarchy Through Explicit and Implicit Cues. *CogSci* (42nd Annual Meeting of the Cognitive Science Society).

56. **Fourtassi, A.**, Scheinfeld, I., & Frank, M. C. (2019). The Development of Abstract Concepts in the Children's Early Lexical Networks. *CMCL* (10th Workshop on Cognitive Modeling and Computational Linguistics).

57. **Fourtassi, A.**, Regan, S., & Frank, M. C. (2019). The Role of Continuous Developmental Change in Explaining Discontinuities in Word Learning. *CogSci* (41st Annual Meeting of the Cognitive Science Society).

58. **Fourtassi, A.**, & Dupoux, E. (2019). Phoneme Learning is Influenced by the Taxonomic Similarity of the Semantic Referents. *CogSci* (41st Annual Meeting of the Cognitive Science Society).

59. **Fourtassi, A.**, Bian, Y., & Frank, M. C. (2018). Word Learning as Network Growth: A Cross-Linguistic Analysis. *CogSci* (40th Annual Meeting of the Cognitive Science Society).

60. **Fourtassi, A.**, & Frank, M. C. (2017). Word Identification Under Multimodal Uncertainty. *CogSci* (39th Annual Meeting of the Cognitive Science Society).

61. **Fourtassi, A.**, & Dupoux, E. (2016). The Role of Word-Word Co-Occurrence in Word-Meaning Learning. *CogSci* (38th Annual Meeting of the Cognitive Science Society).

62. **Fourtassi, A.**, & Dupoux, E. (2014). Rudimentary Lexicon and Semantics Help Bootstrap Phoneme Acquisition. *CoNLL* (18th Conference on Computational Natural Language Learning).

63. **Fourtassi, A.**, Schatz, T., Varadarajan, B., & Dupoux, E. (2014). Exploring the Relative Role of Bottom-Up and Top-Down Information in Phoneme Learning. *ACL* (52nd Annual Meeting of the Association for Computational Linguistics).

64. **Fourtassi, A.**, Dunbar, E., & Dupoux, E. (2014). Self-Consistency as an Inductive Bias in Early Language Acquisition. *CogSci* (36th Annual Meeting of the Cognitive Science Society).

65. **Fourtassi, A.**, Borschinger, B., Johnson, M., & Dupoux, E. (2013). Why Is English So Easy to Segment? *CMCL* (The 4th Workshop on Cognitive Modeling and Computational Linguistics).

66. **Fourtassi, A.**, & Dupoux, E. (2013). A Corpus-Based Evaluation Method for Distributional Semantic Models. *ACL-SRW* (51st Annual Meeting of the Association for Computational Linguistics).

67. Jansen, A., et al. (2013). Zero-Resource Speech Technologies and Models of Early Language Acquisition. *ICASSP* (The International Conference on Acoustics, Speech and Signal Processing).

INVITED TALKS & EXPERT MEETINGS (selection)

- CIFAR Horizon Network: "Learning through Conversation" (2026 - ..)
- International Congress of Infant Studies on "ML for Home Audio Recordings" (ICIS, online event, 2025)
- Mila Quebec AI Institute on "NLP in the Era of AI, Cognitive Science, and Societal Transformations" (2024)
- Dagstuhl Workshop: Developmental Machine Learning (2022)
- OECD Workshop: AI and the Productivity of Science (2021)

MENTORSHIP

PhD supervision - primary advisor

- Marianne Schweitzer (2025-present), co-advised with C. François
- Youcef Brahimi (2025-present)
- Dhia Elhak Goumri (2022-2026; *defended*), co-advised with L. Becerra
  Currently, Research Engineer
- Abhishek Agrawal (2022-2025; *defended*), co-advised with B. Favre
  Currently, Postdoctoral Researcher
- Mitja Nikolaus (2020-2023; *defended*), co-advised with L. Prévot
  🏆 *AMU Best PhD Thesis Award (2024)*
  Currently, Postdoctoral Researcher

PhD supervision - secondary advisor

- Mathias Vigouroux (2025-present), co-advised with J. Sackur
- Jing Liu (2024-present), advised by E. Dupoux
- Kübra Bodur (2022-2025, *defended*), advised by C. Meunier
  🏆 *AFCP Best PhD Thesis Award (2026)*
  Currently, Teaching and Research Fellow (ATER)
- Franck Dary (2018-2022, *defended*), co-advised with A. Nasr

🏆 *Top 2 in CMCL Shared Task (Dary, Nasr, & Fourtassi, 2021)*

Currently, Data Scientist

Postdoctoral supervision

- Giulia Rambelli (2026-present), ILCB postdoctoral fellowship

  Incoming, Lecturer at UCL
- Monica Barbir (2024-present), Marie Skłodowska-Curie European Fellowship
- Chiara Mazzocconi (2020-2023), ILCB postdoctoral fellowship

  Currently, Postdoctoral Researcher

Additional supervision

- Supervised 14 Master's research internships (10 at the M2 level and 4 at the M1 level)
- Supervised 17 summer interns (from Stanford, Jiao Tong, UPenn, ENS Paris, ENS Lyon, Ecole Centrale Marseille, Aix-Marseille)

TEACHING

- Building Minds: The Programming Tools - Master in Cognitive Science MASCO (2025-present)
- Machine Learning for Cognitive Science - MASCO / Master in Computer Science (2024-present; with T. Schatz)
- Data Science (L1 Computer Science, 2025-present)
- Theories in Cognitive Science (MASCO, 2020-2022)
- Introduction to Machine Learning (L3 Computer Science, 2022-2023)
- Introduction to Natural Language Processing (M1 Computer Science, 2020-2023)
- Probability Theory for Computer Science (M1, 2022)
- ILCB Summer School: week-long course on deep learning and language development (2025, 2026)
- ETAL National Summer School on NLP: When NLP Meets Cognitive Science (2023)

EDITORIAL, REVIEWING & SERVICE

Editorial roles in Journals

- Associate Editor, *Dialogue & Discourse* (2026-present)
- Associate Editor, *Frontiers in Cognitive Neuroscience* (2023-2025)
- Guest Editor, *Computational Linguistics* (2024), with M. Apidianaki (UPenn) and S. Pado (U. of Stuttgart)

Editorial roles in Conferences

- Program Chair, *1st Workshop on Computational Developmental Linguistics* at ACL (2026)
- Senior Area Chair, *COLING* (2025, 2027), *ACL* (2027)
- Area Chair, *ACL Rolling Review* (2025-present)
- Meta-Reviewer, *CogSci* (2026)

Reviewing

- Funding agencies: ANR (France), NSF (USA), JST (Japan), NWO (Netherlands), British Academy (UK), Royal Society (UK)
- Selected journals: Psychological Review, Trends in Cognitive Sciences, Proceedings B, Cognition, Cognitive Science, Child Development, Developmental Psychology, JEP: General, Behavior Research Methods
- Selected conferences: ACL Rolling Review, CogSci, ICMI, CoNLL, CDS, ACII, CMCL

Service & Events (recent)

- Co-organizer, workshop on *Developmental AI,* CNRS GDR TAL / Babylab, 2026 (with M. Fort, T. Hueber)
- Co-organizer, First workshop on *Computational Developmental Linguistics,* at *ACL* 2026
- Selection Committee panel, France-Japan bilateral funding, ANR-JST CREST, 2026
- Co-organizer, CIVIS Europe-wide collaborative workshop, 2026 (with M. Jover)
- Co-organizer, national workshop on *NLP & Cognition*, CNRS GDR TAL, 2025 (with P. Denis, L. Prévot)
- Hiring panel, Associate Professor position in Computer Science, Université de Lille, 2025
- Member of 13 thesis advisory committees

PUBLIC OUTREACH

- Chapter "What AI Tells Us About Language Development" in the popular science book: *Parole de Neurones* (forthcoming)
- Bringing the lab to the classroom, ANR-funded Festival Explore (2026)
- Interview: AI and Cognitive Science, Photizo podcast (2025)
- Interactive MACoMiC project stand, Nuit des chercheurs (2023)
- Article in *The Conversation:* "Infants learn the art of conversation before they speak" (2022), with R. Bertrand and N. Nguyen